

GENX: Porewater Methane

April Samples

2026-06-11

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##Run Information

```
cat("Run Information: Melanie Giessner ") #lets you know what section you're in
```

Run Information: Melanie Giessner

```
#set the run date & user name
sample_year <- "2026"
sample_month <- "April"
user <- "Melanie Giessner"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

#change dates
folder_path_rawdata <- paste0("Raw Data")
GHG_files <- list.files(path = folder_path_rawdata,pattern = "20260417|20260501" ,full.names = TRUE)
head(GHG_files)
```

```
## [1] "Raw Data/20260417_GENX_GHGdata.csv" "Raw Data/20260501_GENX_GHGdata.csv"
```

```
#GHG_files <- c("Raw Data/20260417_GENX_GHGdata.csv","Raw Data/20260501_GENX_GHGdata.csv" )

# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/ShimadzuGC_Slope_QAQC.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_GENX_Porewater_CH4_202604.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)

volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)

#record any notes about the run or anything other info here:
run_notes <- ""

#Metadata File Path
Raw_Metadata = here("..", "GCReW", "GCReW_Project_Treatment_Metadata.csv")

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

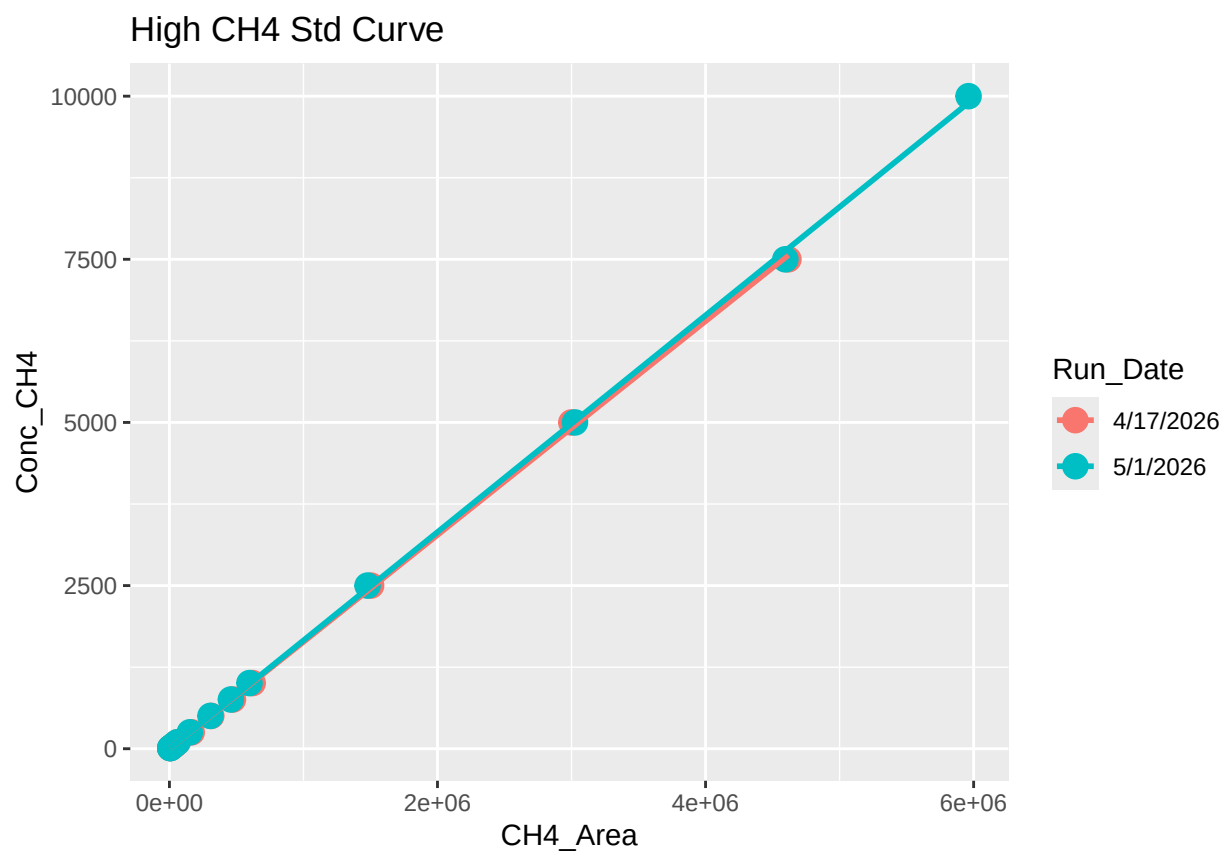
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

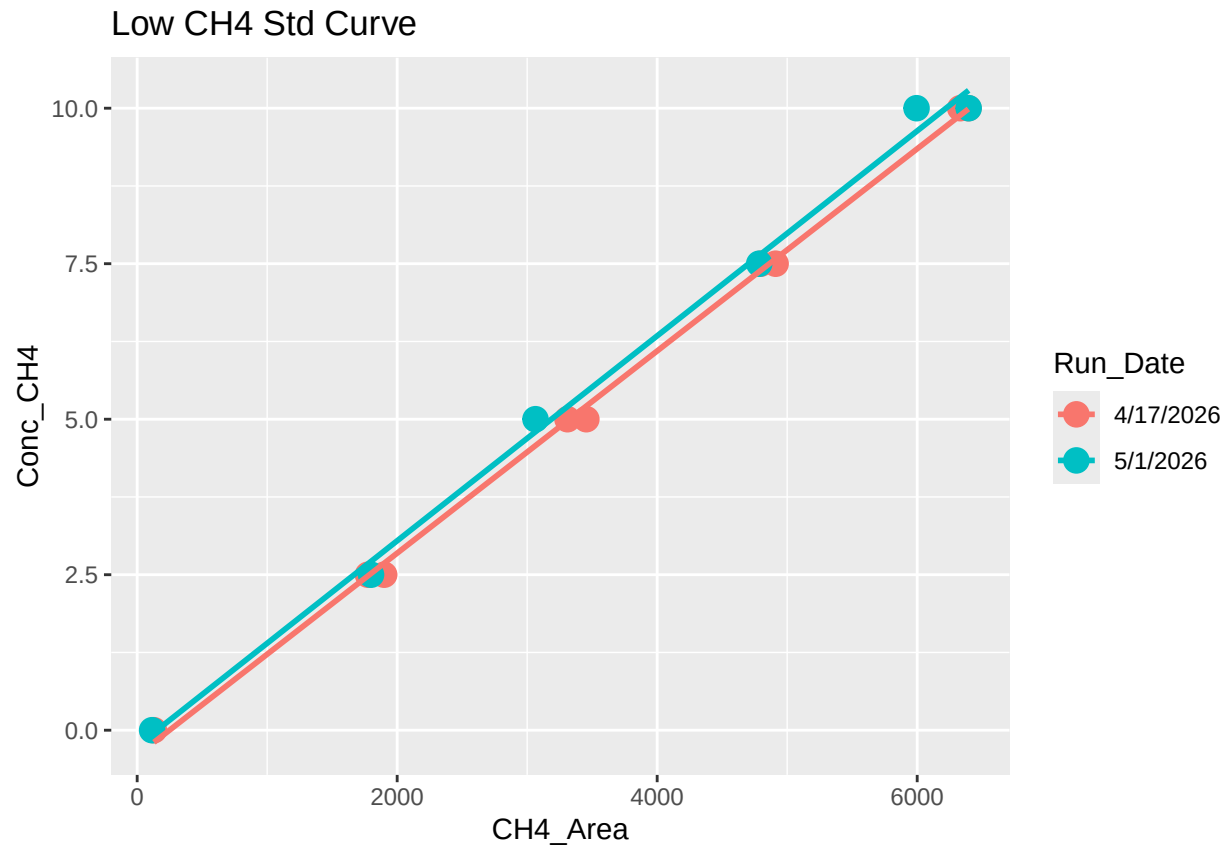
1 Read in data files

2 Assess standard curves

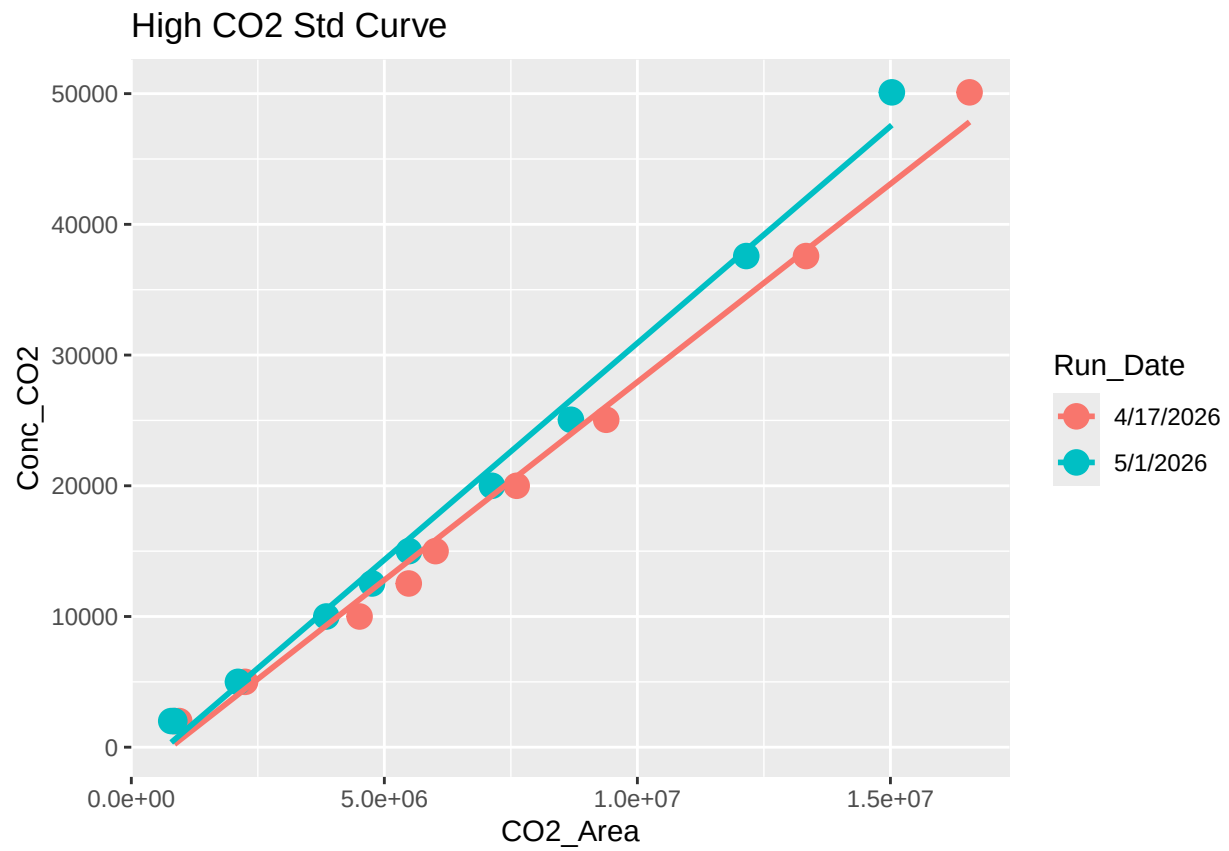
```
## 'geom_smooth()' using formula = 'y ~ x'
```



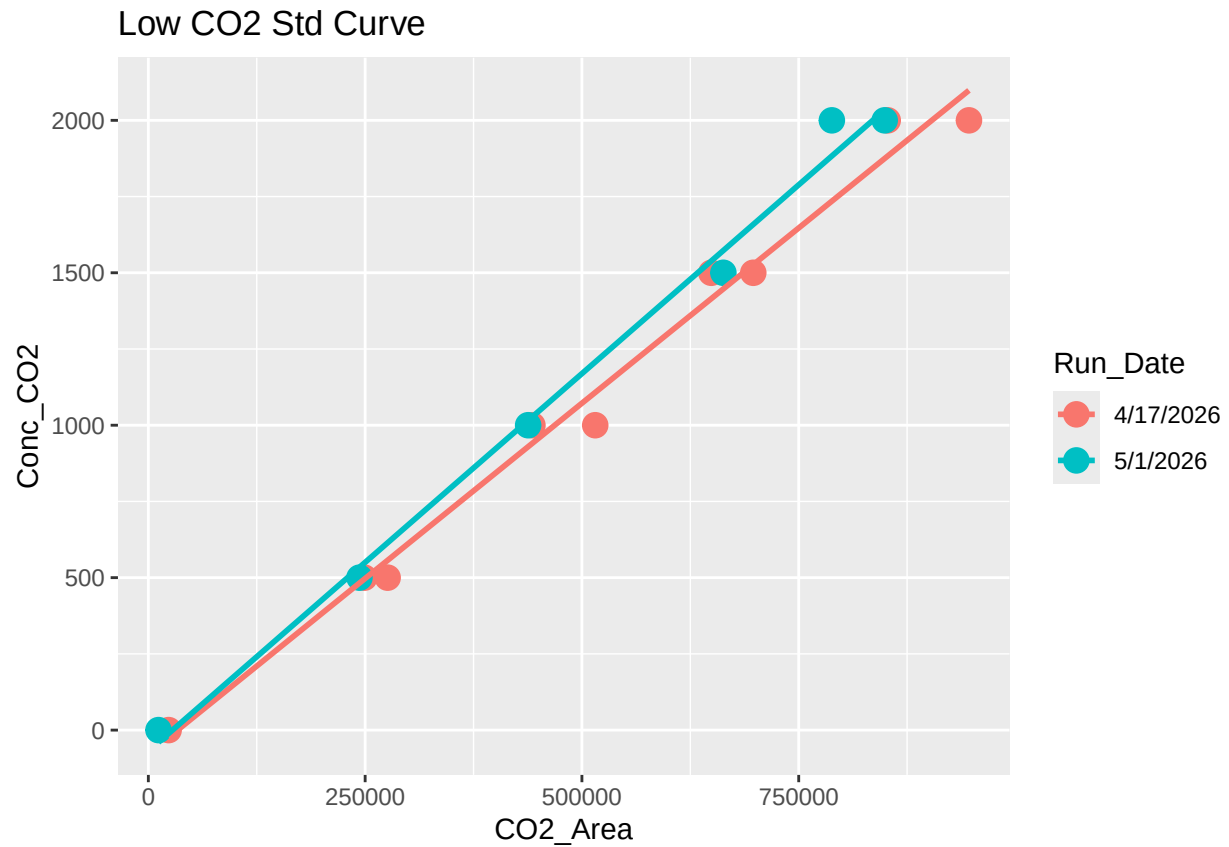
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Joining with 'by = join_by(Run_Date)'  
  
## [1] "High_R2 is Good-4/17/2026" "High_R2 is Good-5/1/2026"  
  
## [1] "Low_R2 is Good-4/17/2026" "Low_R2 is Good-5/1/2026"  
  
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

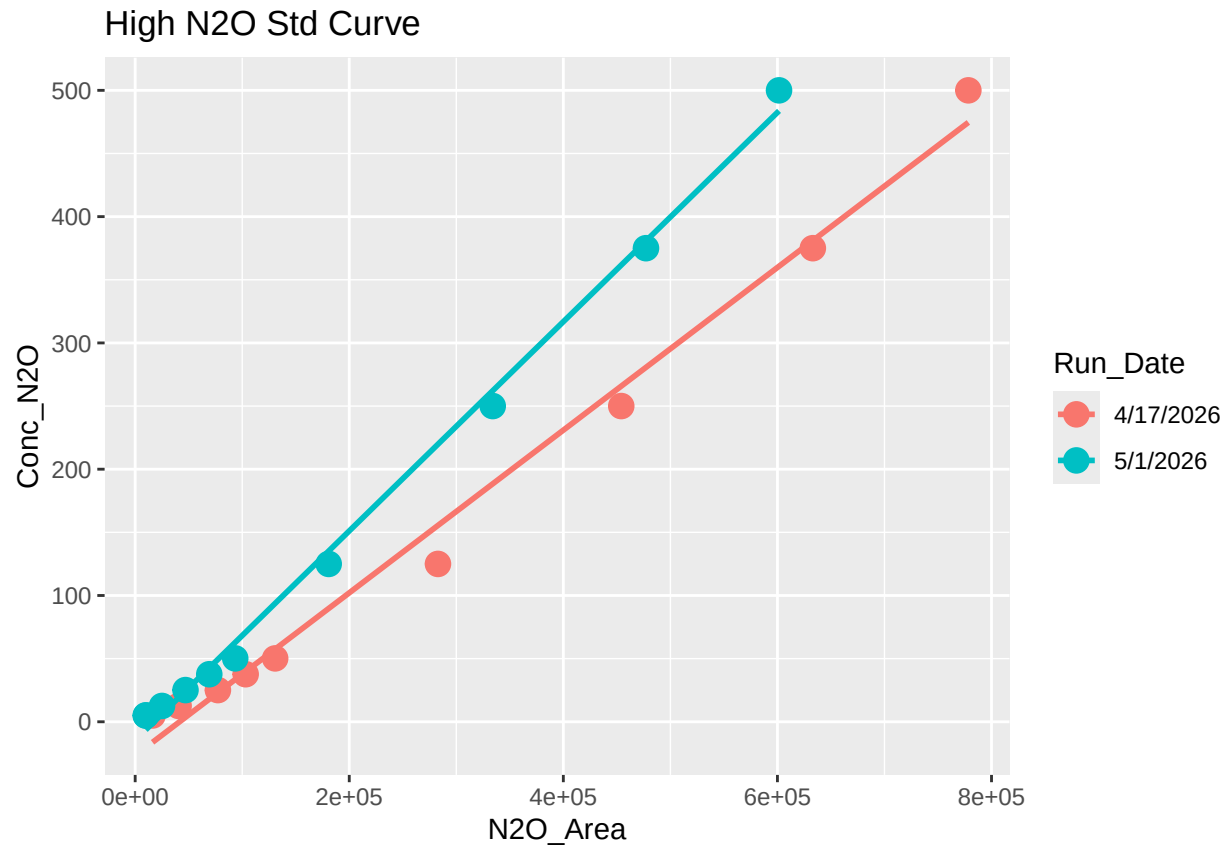


```
## Joining with 'by = join_by(Run_Date)'

## [1] "High_R2 is Good-4/17/2026" "High_R2 is Good-5/1/2026"

## [1] "Low_R2 is Bad-4/17/2026" "Low_R2 is Good-5/1/2026"

## 'geom_smooth()' using formula = 'y ~ x'
```



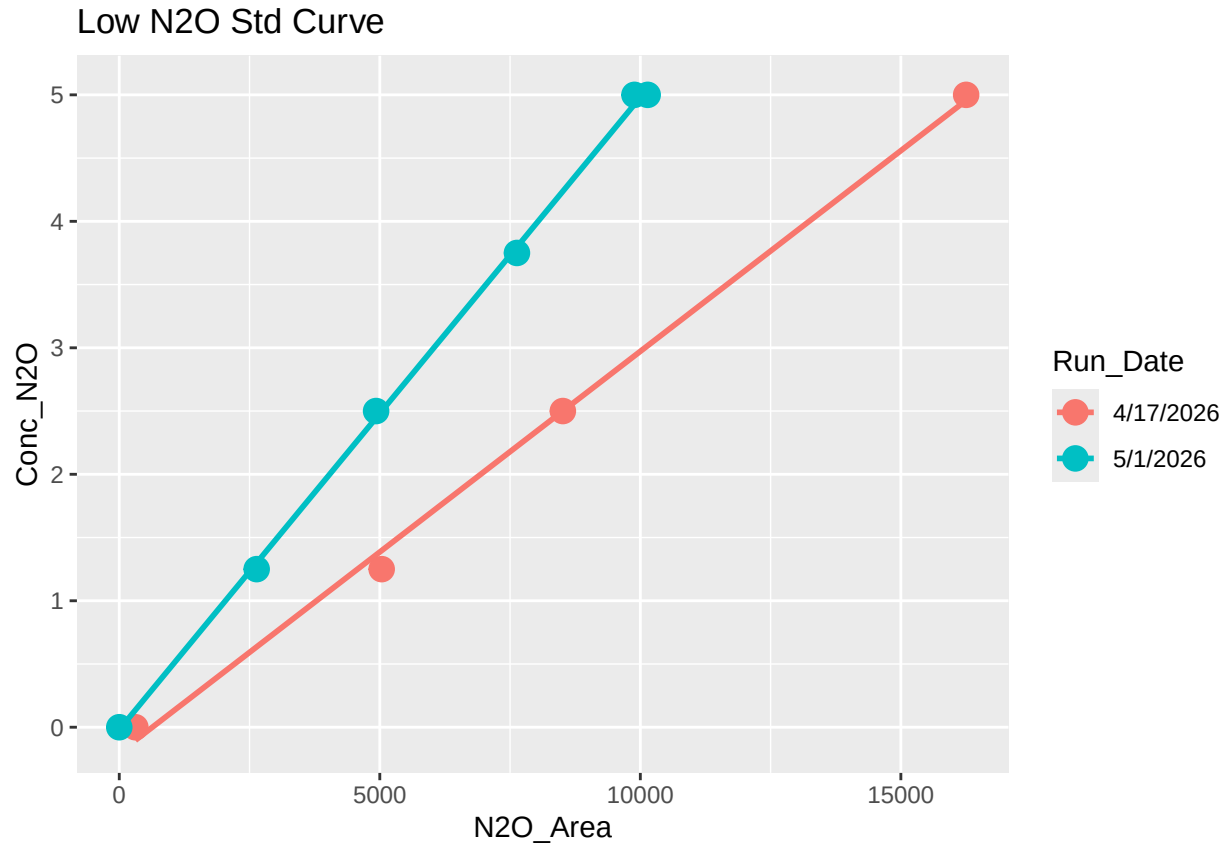
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
```

```
## ('geom_point()').
```



```
## Joining with 'by = join_by(Run_Date)'
```

```
## [1] "High_R2 is Bad-4/17/2026" "High_R2 is Good-5/1/2026"
```

```
## [1] "Low_R2 is Good-4/17/2026" "Low_R2 is Good-5/1/2026"
```

```
#add new standard curve data to log
```

```
Slope_CH4_1 <- Slope_CH4 %>% mutate(curve = paste0("Slope_CH4_", CH4_Curve), R2 = CH4_R2, slope = CH4_S
```

```
CH4_Curve == "Low" ~ paste(CH4_low_std_crv_cutoff,"ppm"),
```

```
CH4_Curve == "High" ~ paste(CH4_high_std_crv_cutoff,"ppm"),
```

```
.default = as.character(CH4_mid_std_crv_cutoff)
```

```
)) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
```

```
Slope_CO2_1 <- Slope_CO2 %>% mutate(curve = paste0("Slope_CO2_", CO2_Curve), R2 = CO2_R2, slope = CO2_S
```

```
CO2_Curve == "Low" ~ paste(CO2_low_std_crv_cutoff,"ppm"),
```

```
CO2_Curve == "High" ~ paste(CO2_high_std_crv_cutoff,"ppm"),
```

```
.default = as.character(CO2_mid_std_crv_cutoff)
```

```
)) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
```

```
Slope_N2O_1 <- Slope_N2O %>% mutate(curve = paste0("Slope_N2O_", N2O_Curve), R2 = N2O_R2, slope = N2O_S
```

```
N2O_Curve == "Low" ~ paste(N2O_low_std_crv_cutoff,"ppm"),
```

```
N2O_Curve == "High" ~ paste(N2O_high_std_crv_cutoff,"ppm"),
```

```
.default = as.character(N2O_mid_std_crv_cutoff)
```

```
)) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
```

```
Slope <- rbind(Slope_CH4_1, Slope_CO2_1, Slope_N2O_1)
```



```
log <- read.csv(log_path)
log <- log[, -c(1)]
head(log)
```

```
## Analysis Run_Date Run_by Curve_top curve slope
## 1 CH4 2024-07-09 Alia Al-Haj 10 ppm Slope_Ch4_low 0.001594508
## 2 CH4 2024-07-09 Alia Al-Haj 100 ppm Slope_Ch4_high 0.001562326
## 3 N2O 2024-07-09 Alia Al-Haj 5 ppm Slope_N2O_low 0.000192023
## 4 N2O 2024-07-09 Alia Al-Haj 50.2 ppm Slope_N2O_high 0.000282245
## 5 CO2 2024-07-09 Alia Al-Haj 2000 ppm Slope_CO2_low 0.000282245
## 6 CO2 2024-07-09 Alia Al-Haj 20000 ppm Slope_CO2_high 0.001549502
## intercept R2
## 1 -0.20901179 0.9960215
## 2 0.09646243 0.9996531
## 3 0.07848462 0.9940785
## 4 -0.49925490 0.9960644
## 5 -0.49925490 NA
## 6 38.64567509 0.9996907
```

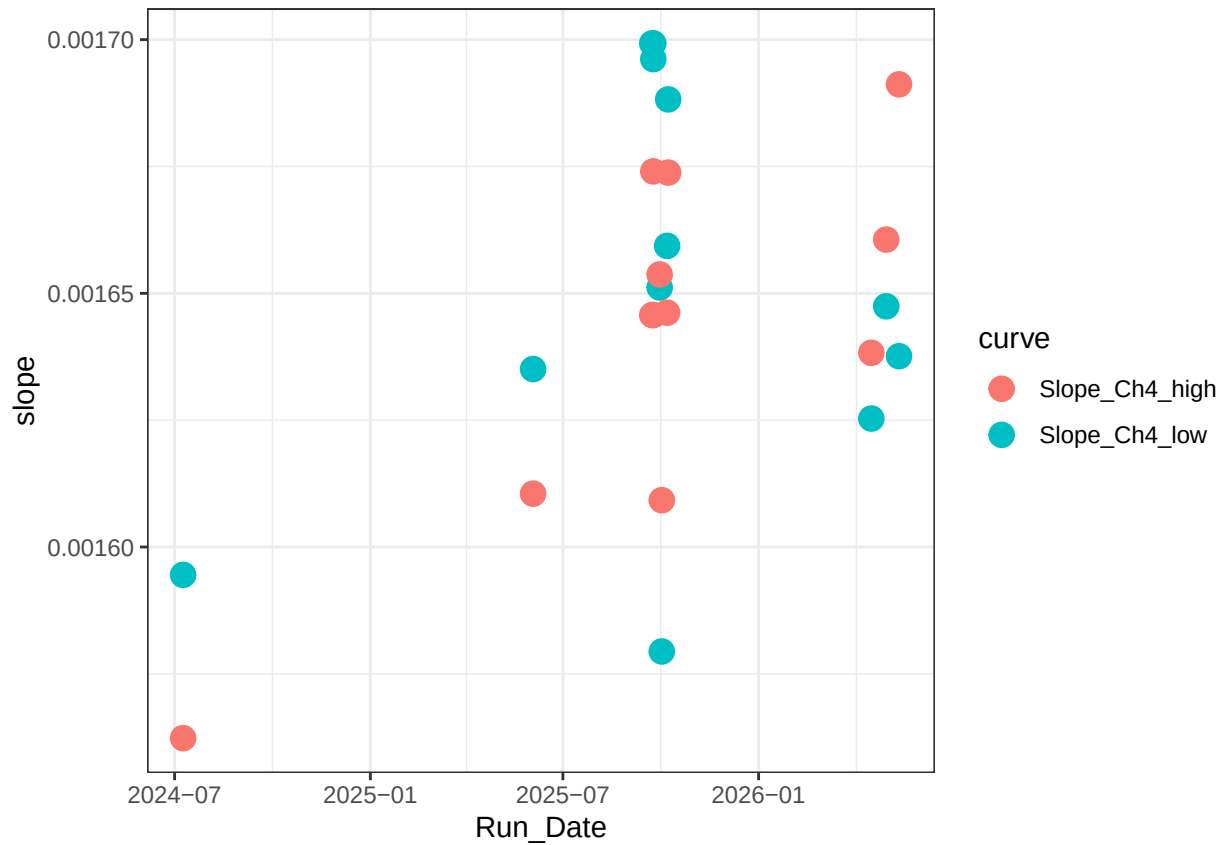
```
log <- rbind(log, Slope)
```

#remove possible duplicates from multiple QAQC attempts

```
unique_log <- log %>%
  mutate(Run_Date = as.Date(Run_Date, tryFormats = c("%Y-%m-%d", "%m/%d/%Y", "%m/%d/%y")))
Slopes_chk_CH4 <- ggplot(unique_log %>% filter(grepl("CH4", curve, ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw())
```

```
Slopes_chk_CH4
```

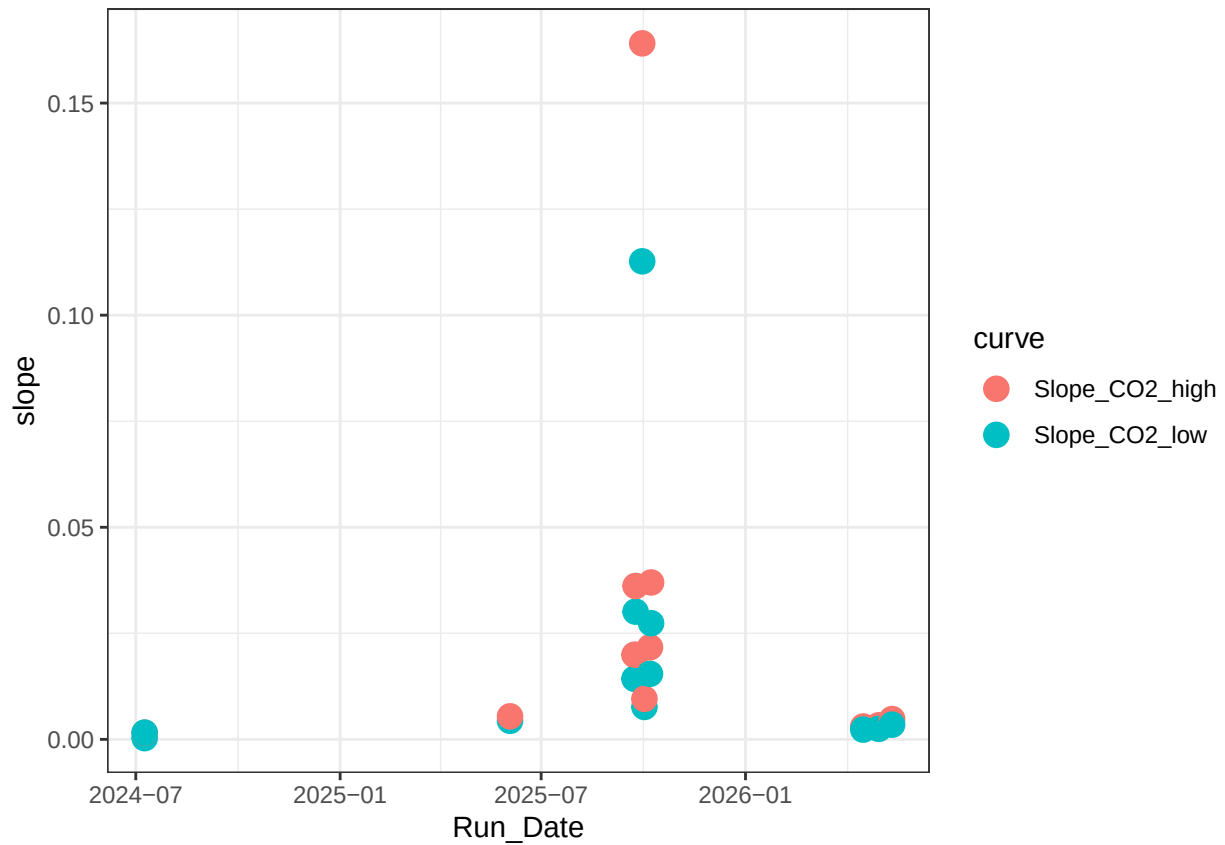
```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
Slopes_chk_CO2 <- ggplot(unique_log %>% filter( grepl("CO2",curve,ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

Slopes_chk_CO2

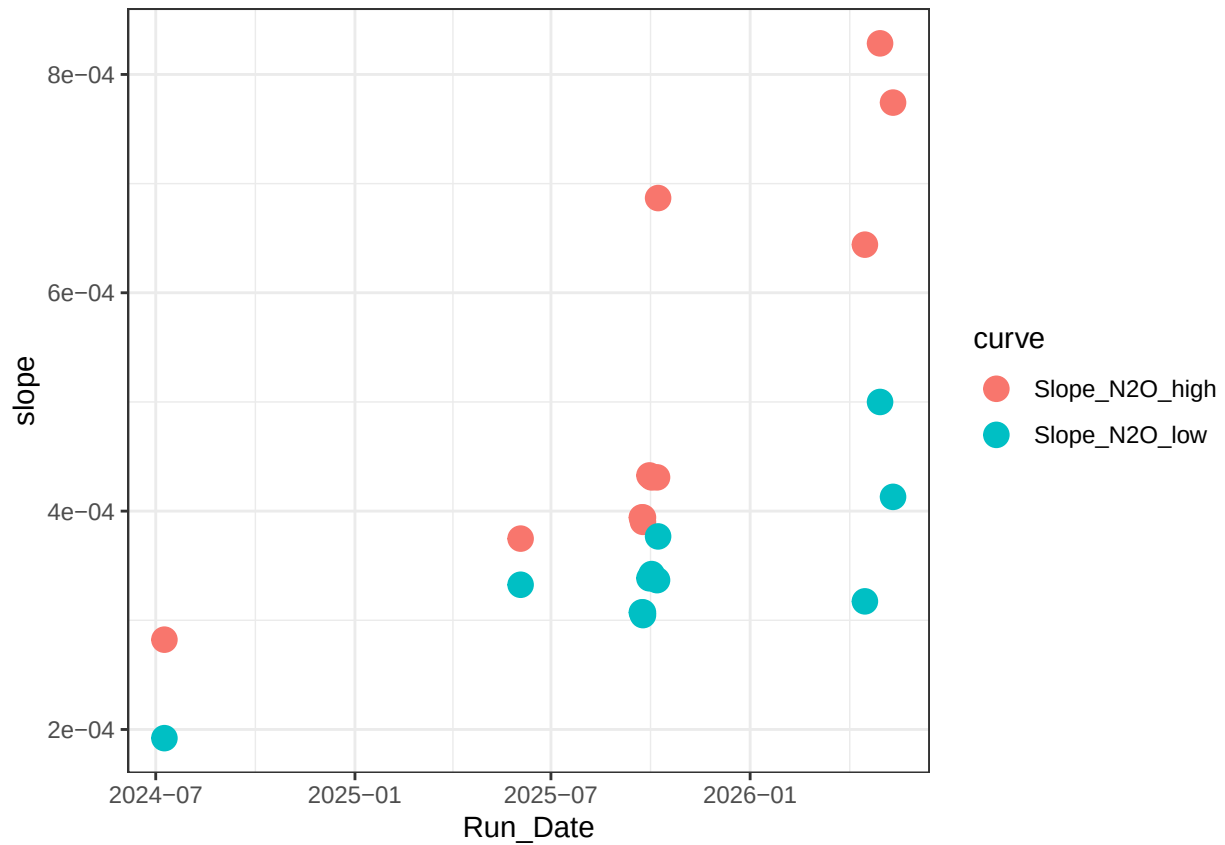
```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
Slopes_chk_N20 <- ggplot(unique_log %>% filter( grepl("N20",curve,ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

Slopes_chk_N20

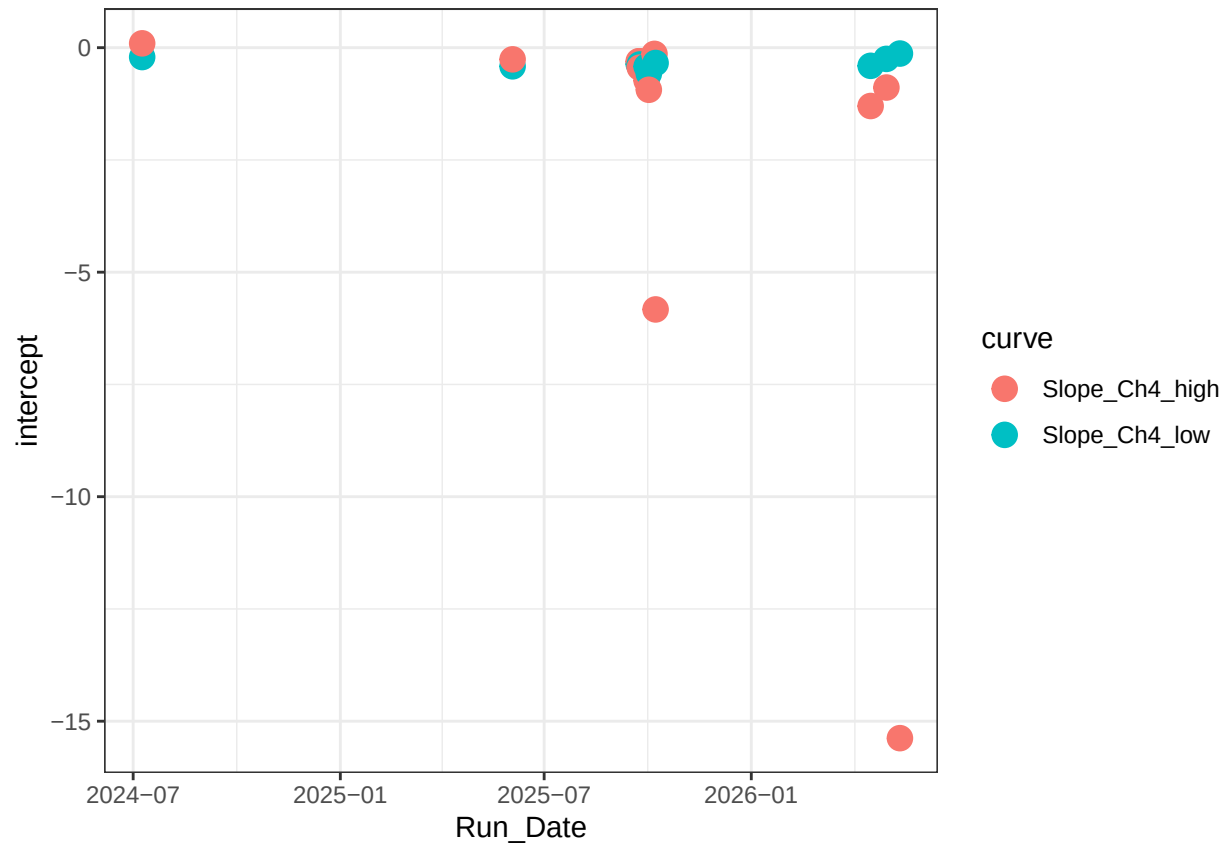
```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
int_chk_CH4 <- ggplot(unique_log %>% filter( grepl("CH4",curve,ignore.case = T)), aes(x = Run_Date, y = slope)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

```
int_chk_CH4
```

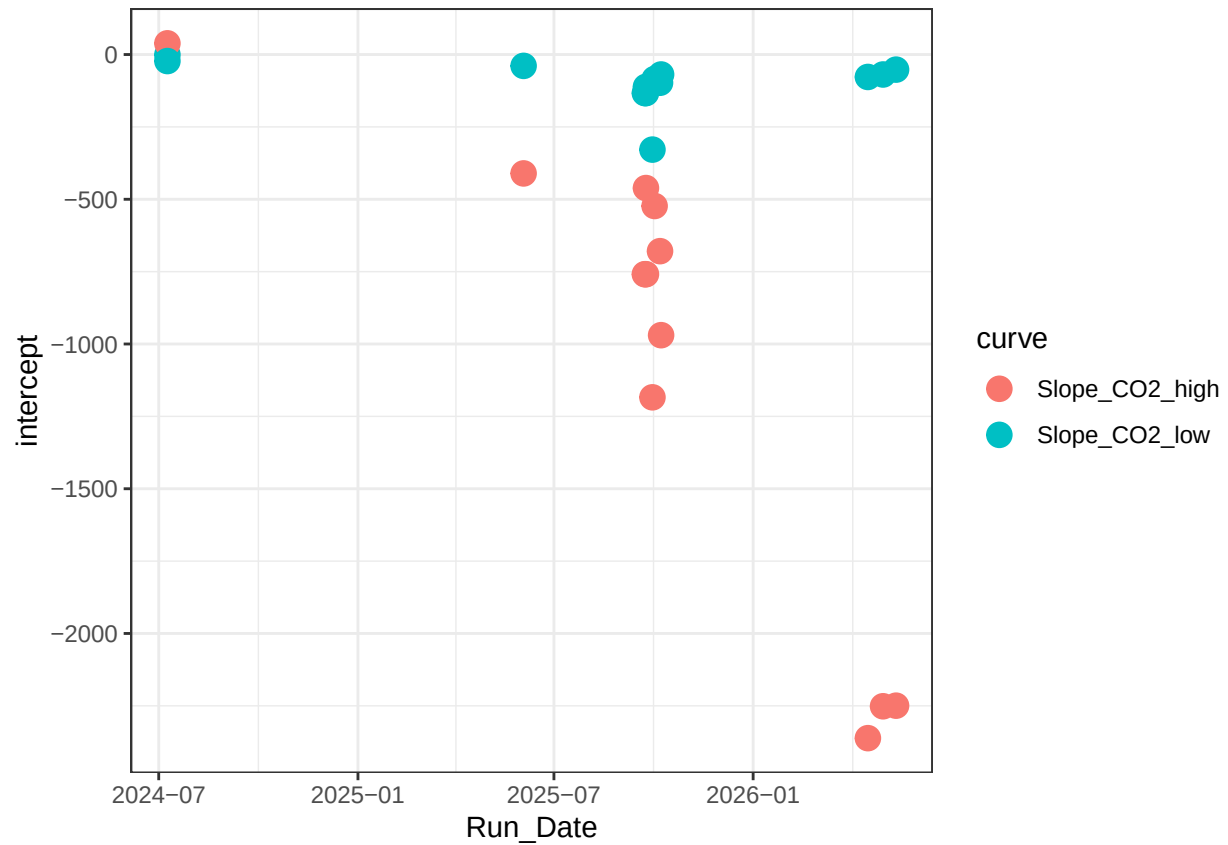
```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
int_chk_C02 <- ggplot(unique_log %>% filter( grepl("C02",curve,ignore.case = T)), aes(x = Run_Date, y = intercept)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

```
int_chk_C02
```

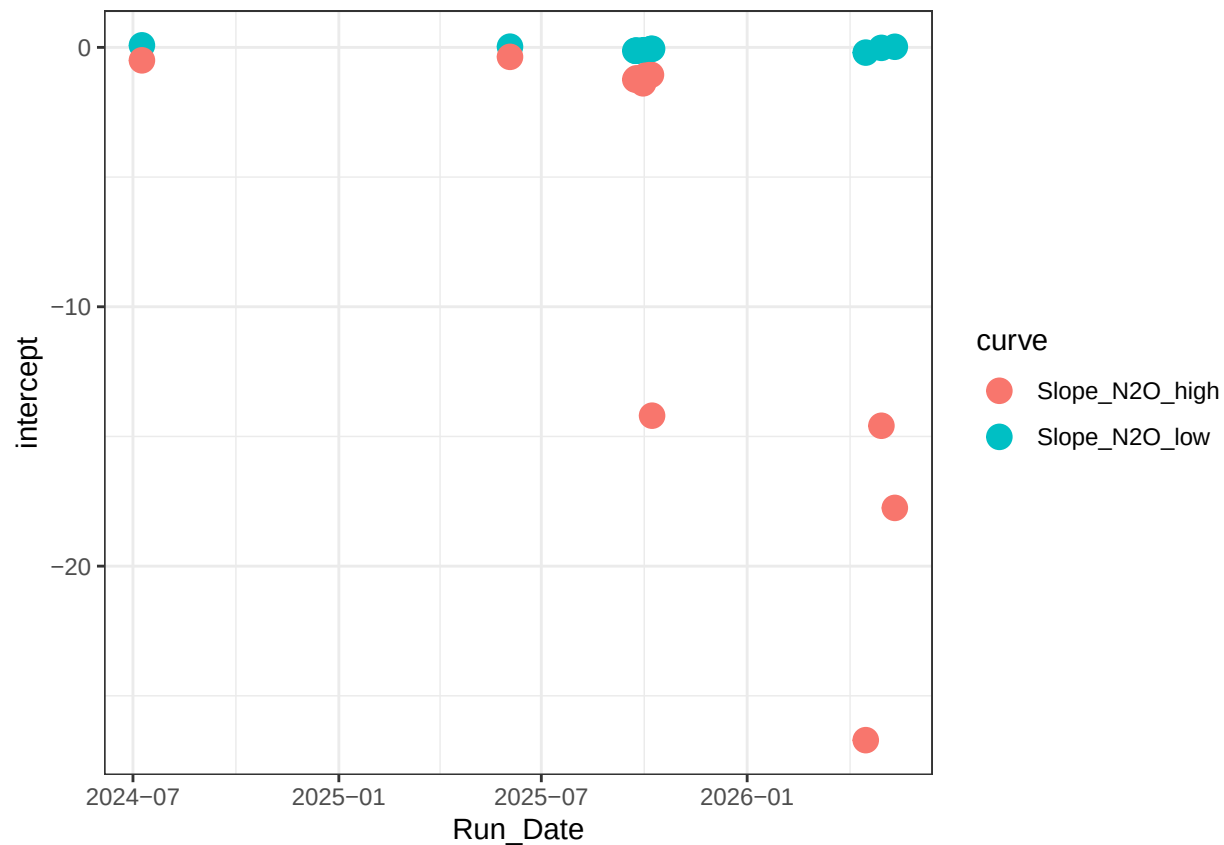
```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
int_chk_N20 <- ggplot(unique_log %>% filter( grepl("N20",curve,ignore.case = T)), aes(x = Run_Date, y = intercept)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

```
int_chk_N20
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').
```



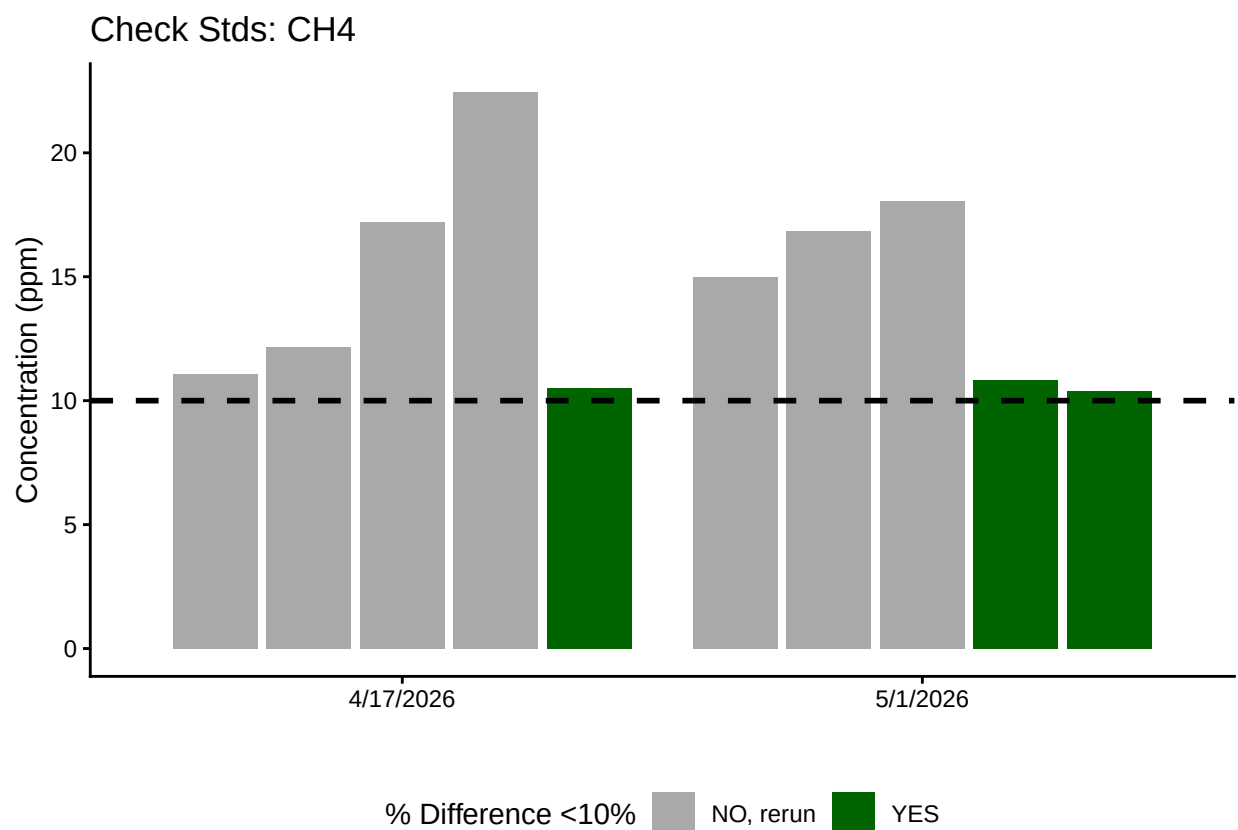
2.1 Now calculate the CH4 in ppm

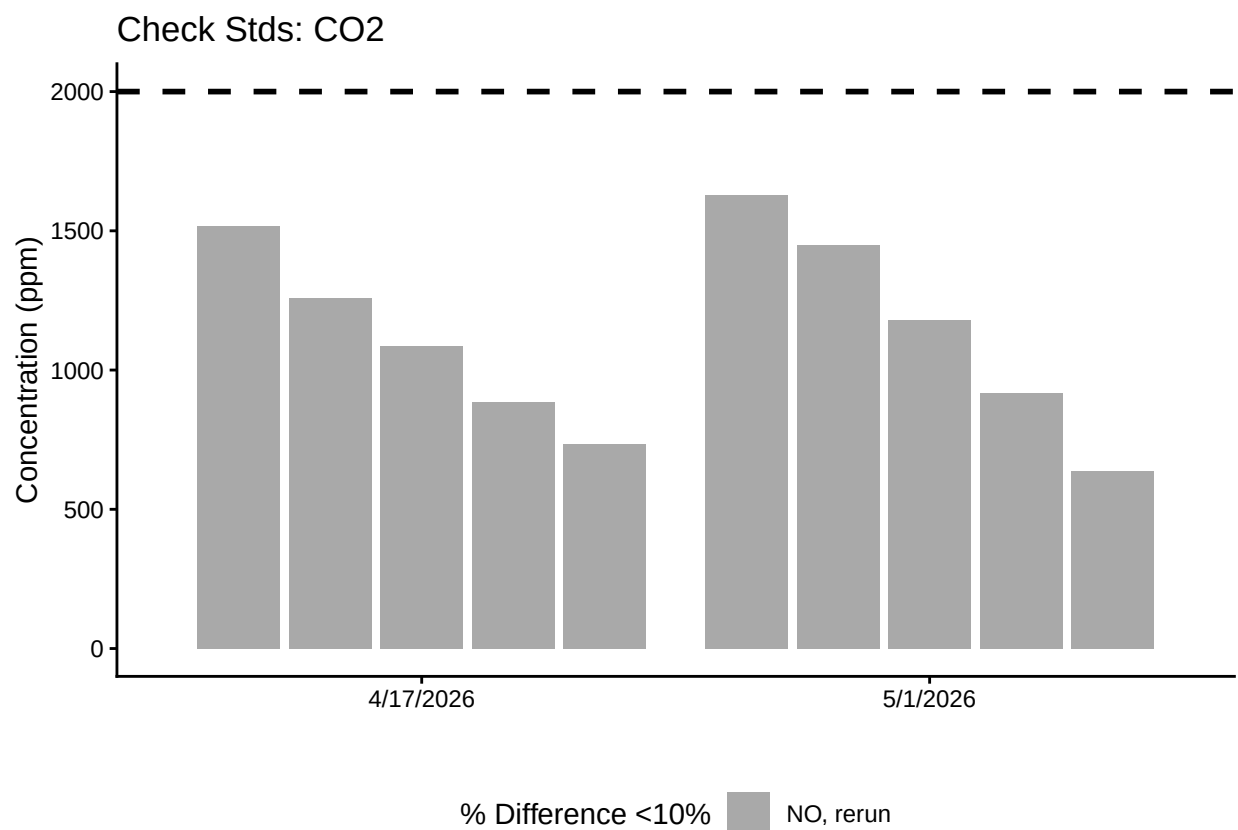
3 Aanlayze the Check Standards

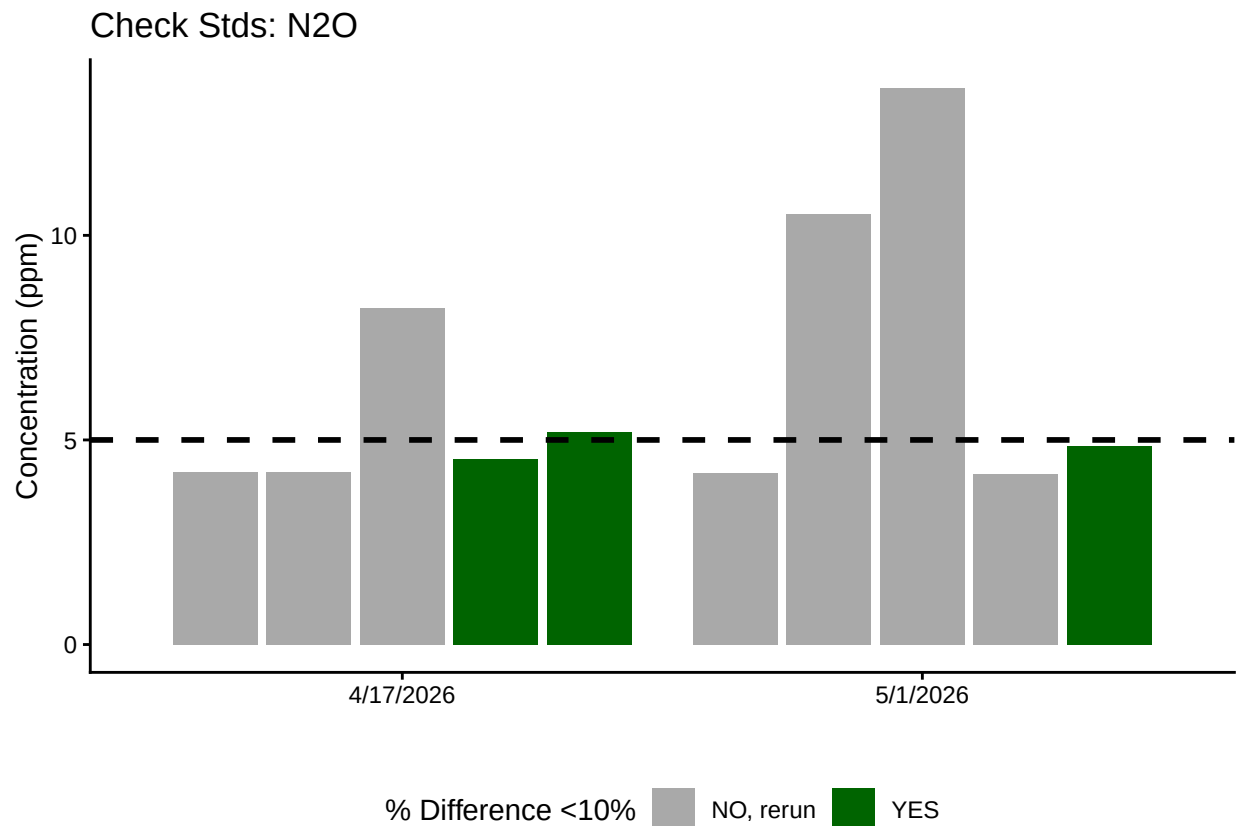
<60% of CH4 Check Standards are within range of expected concentration - REASSESS

<60% of CO2 Check Standards are within range of expected concentration - REASSESS

<60% of N2O Check Standards are within range of expected concentration - REASSESS







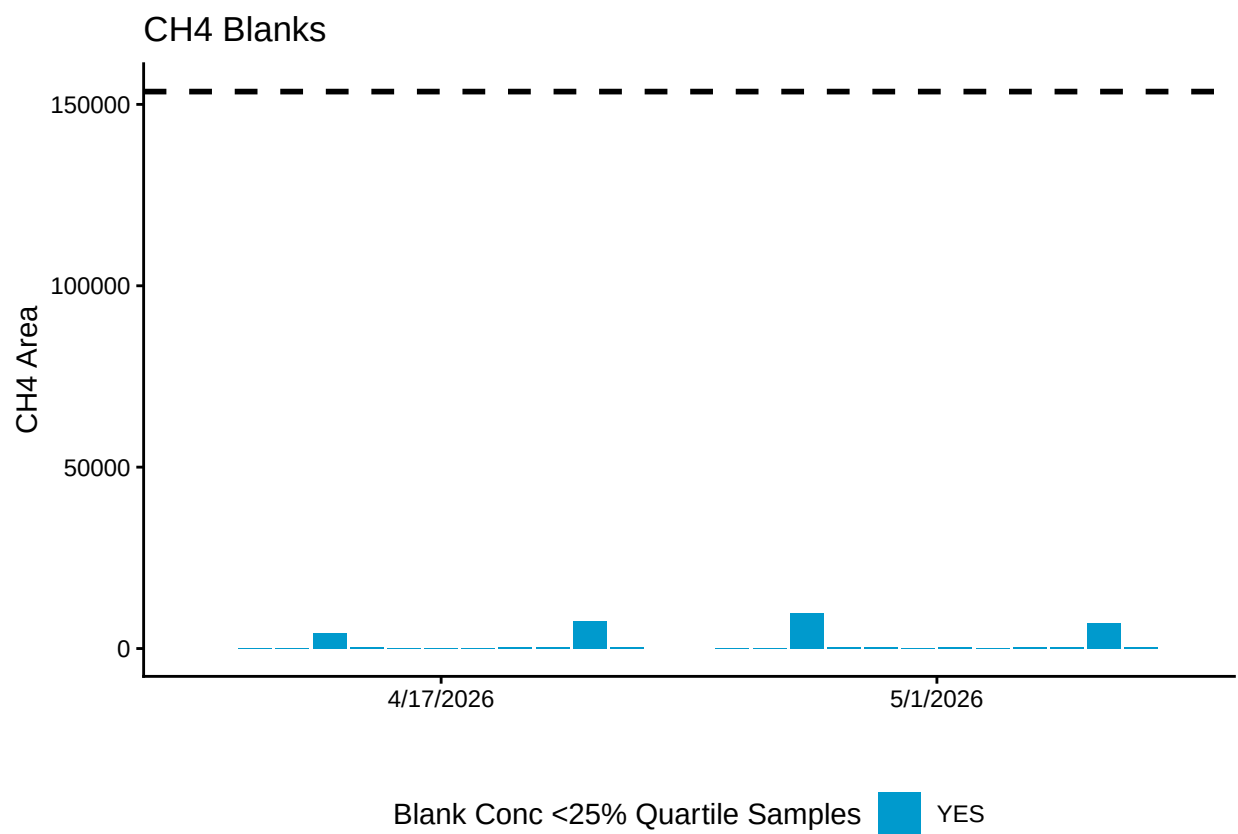
4 Analyze the Blanks

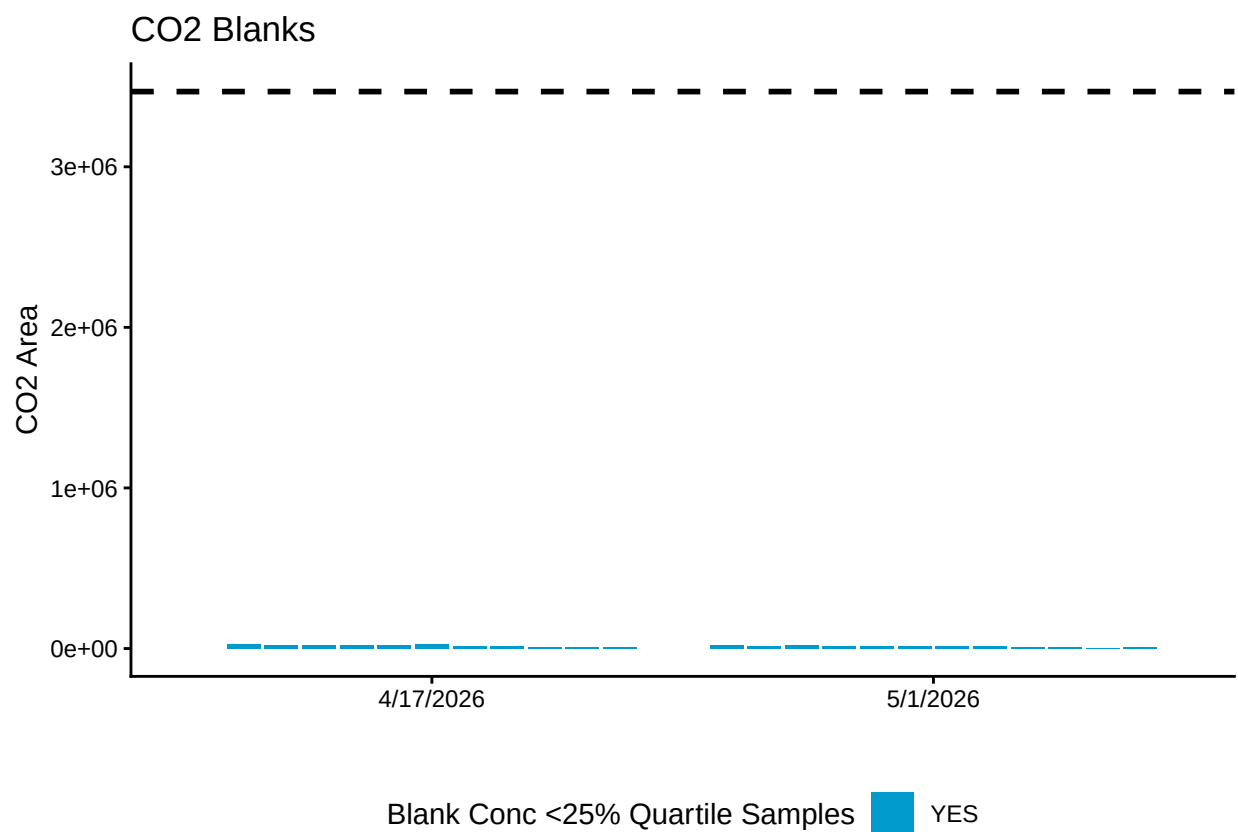
Assess Blanks

>60% of CH₄ Blanks are below the lower 25% quartile of samples - PROCEED

>60% of CO₂ Blanks are below the lower 25% quartile of samples - PROCEED

>60% of N₂O Blanks are below the lower 25% quartile of samples - PROCEED





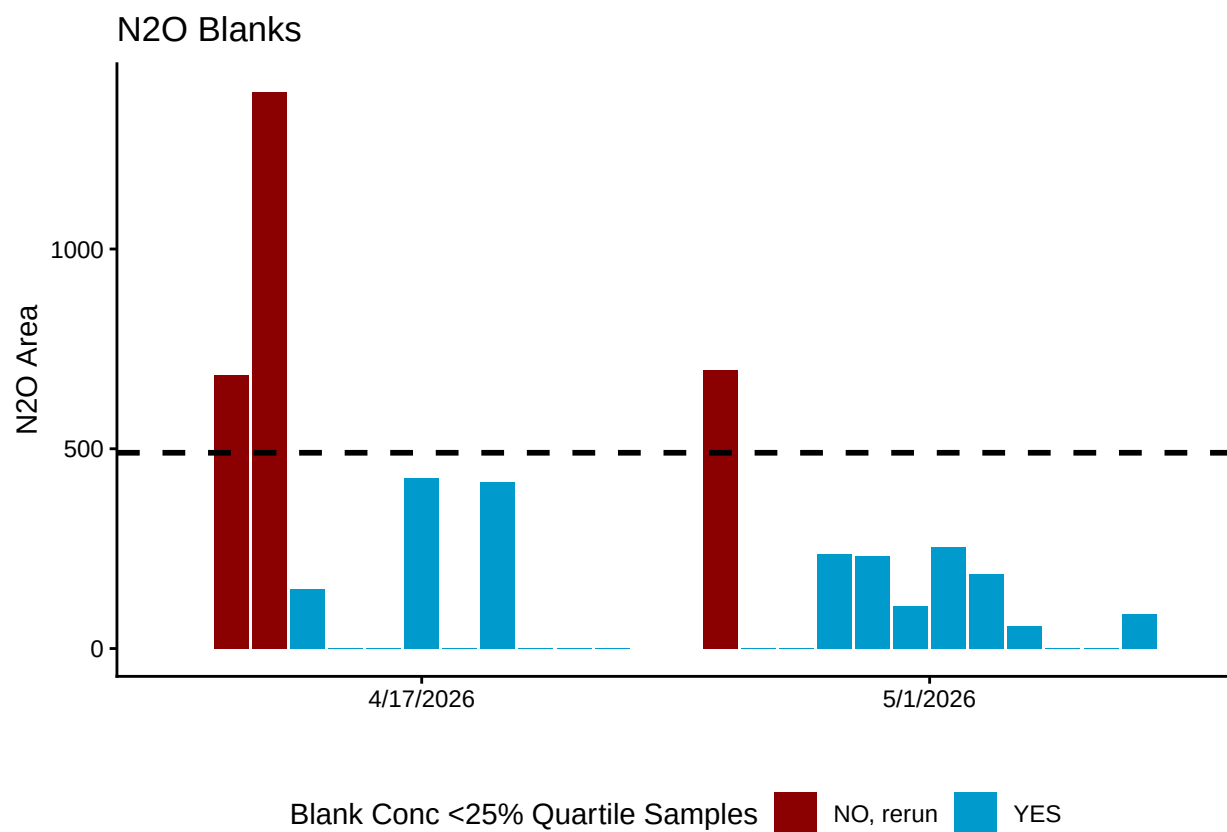
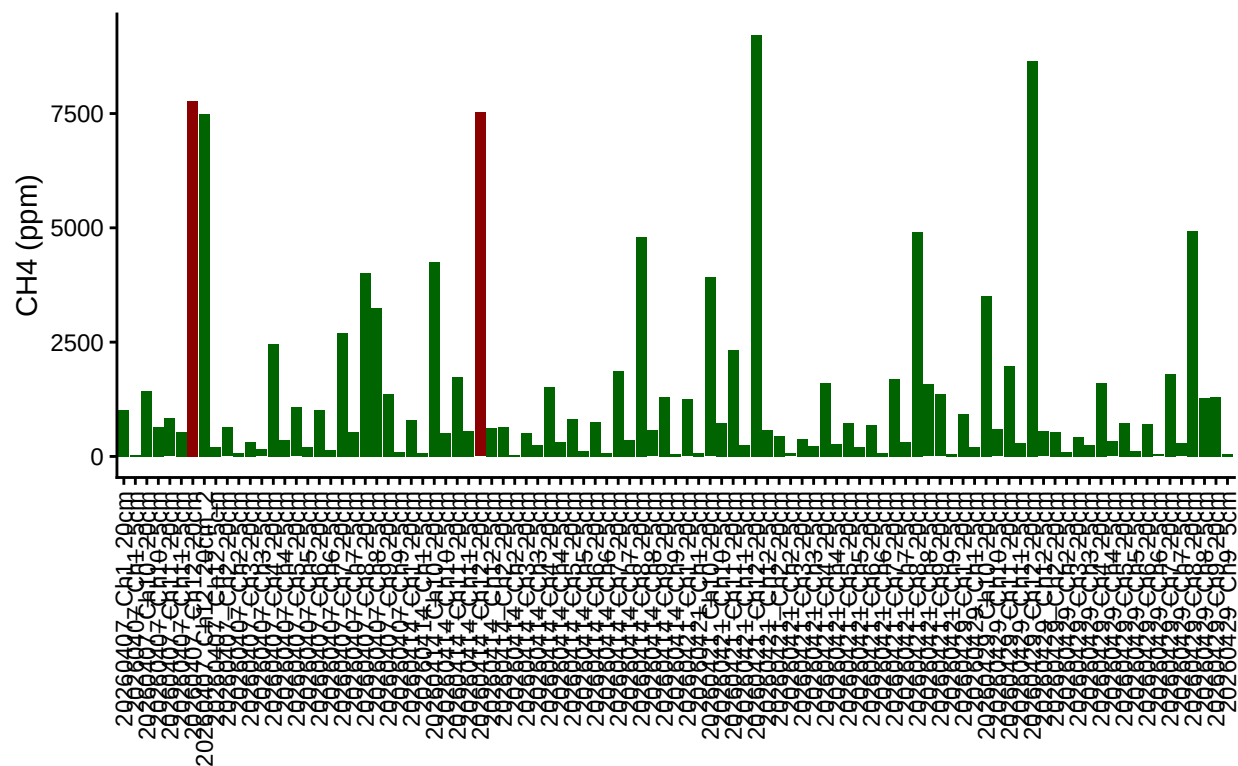


Table 1: Mean Area of Blanks

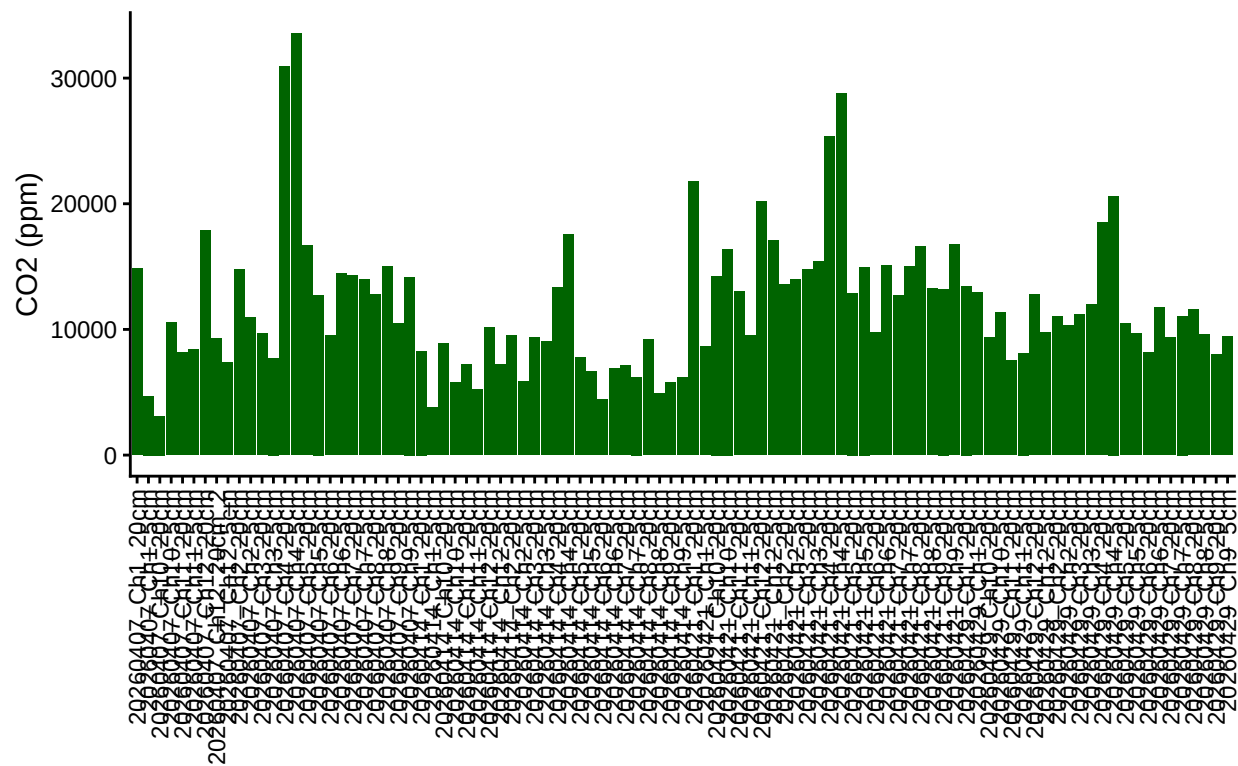
Test	Blank_Mean_Area
CH4	1502.4348
CO2	15675.8696
N2O	214.1304

5 Dilution correct samples and Check Range

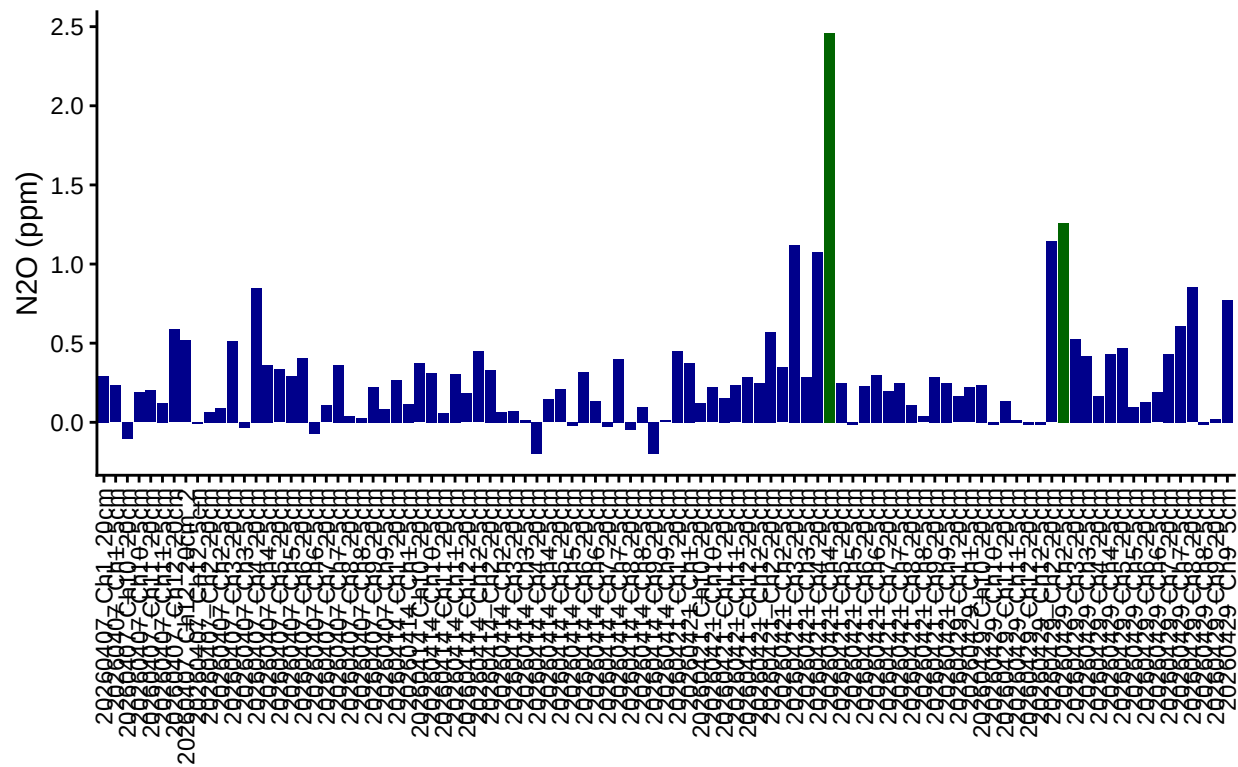
CH4: Green = Within Range (Not Dil Cor)



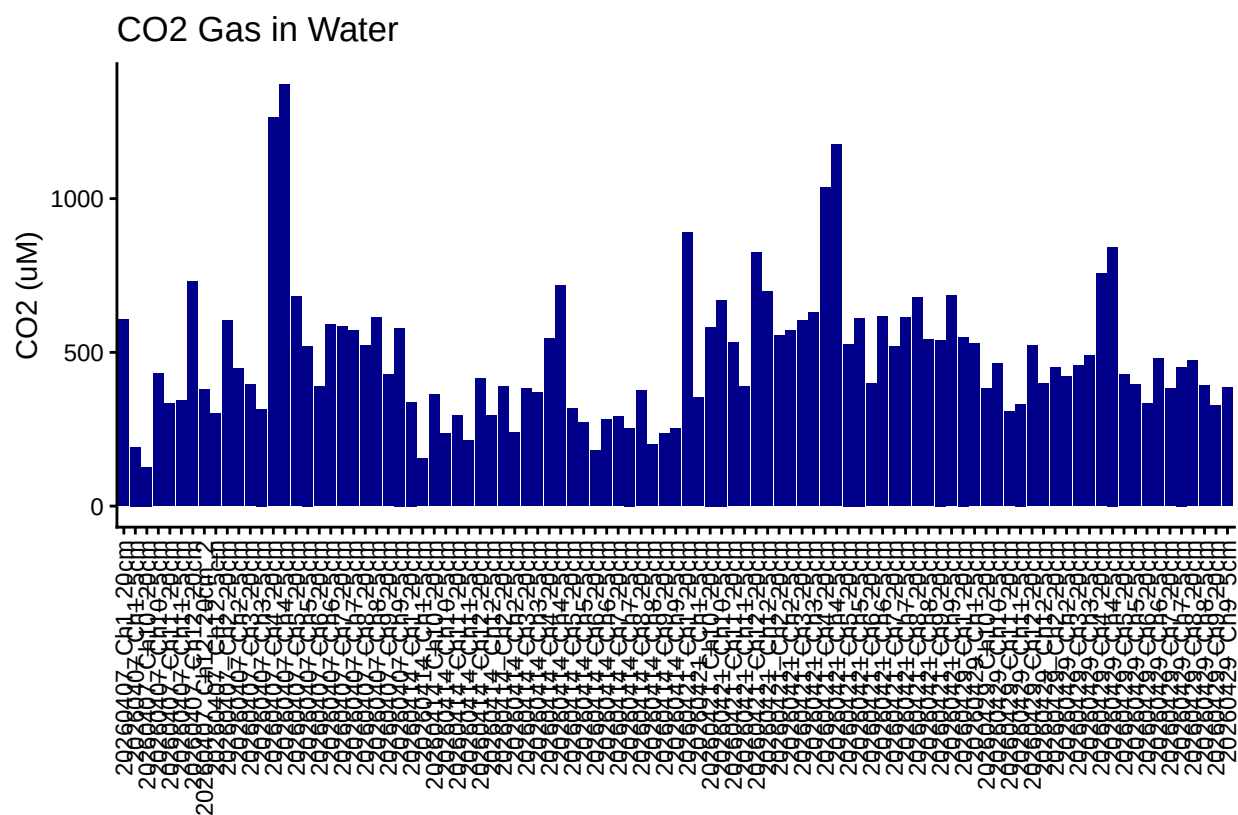
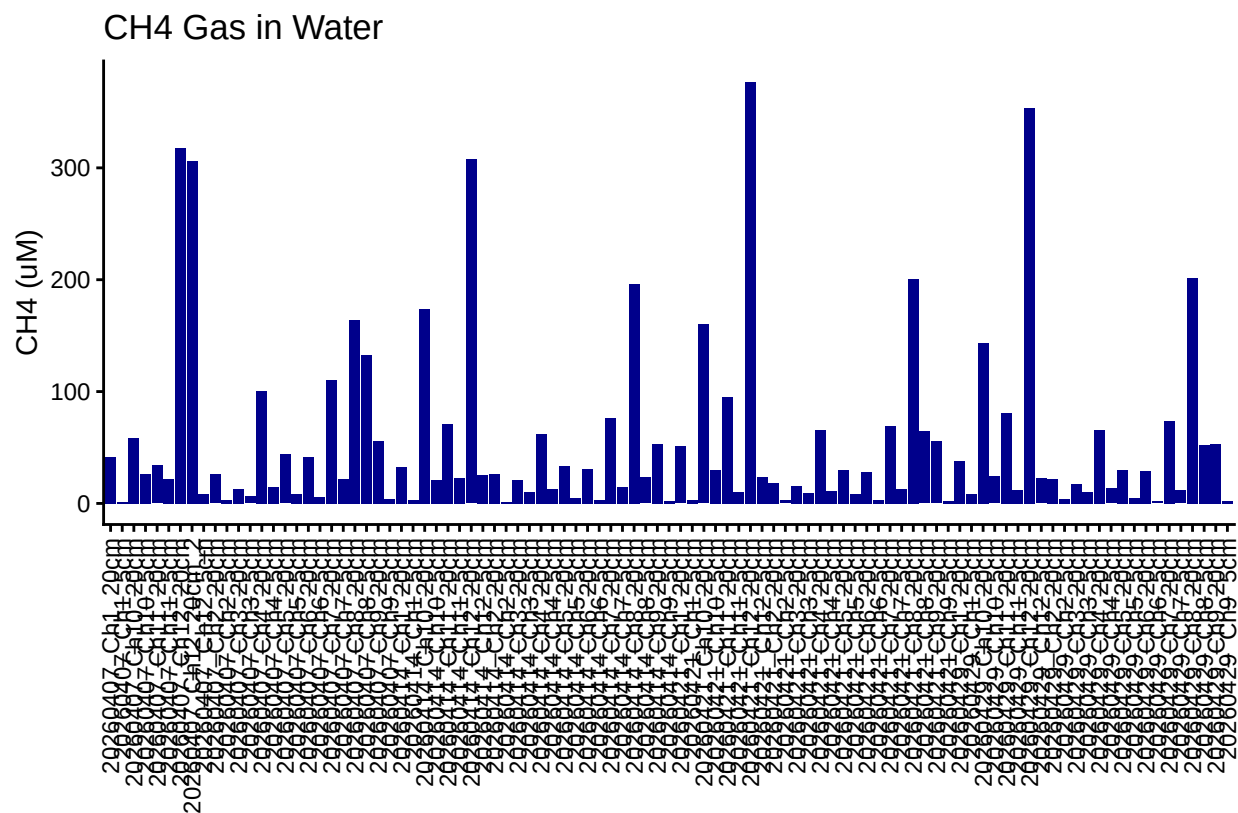
CO2: Green = Within Range (Not Dil Cor)

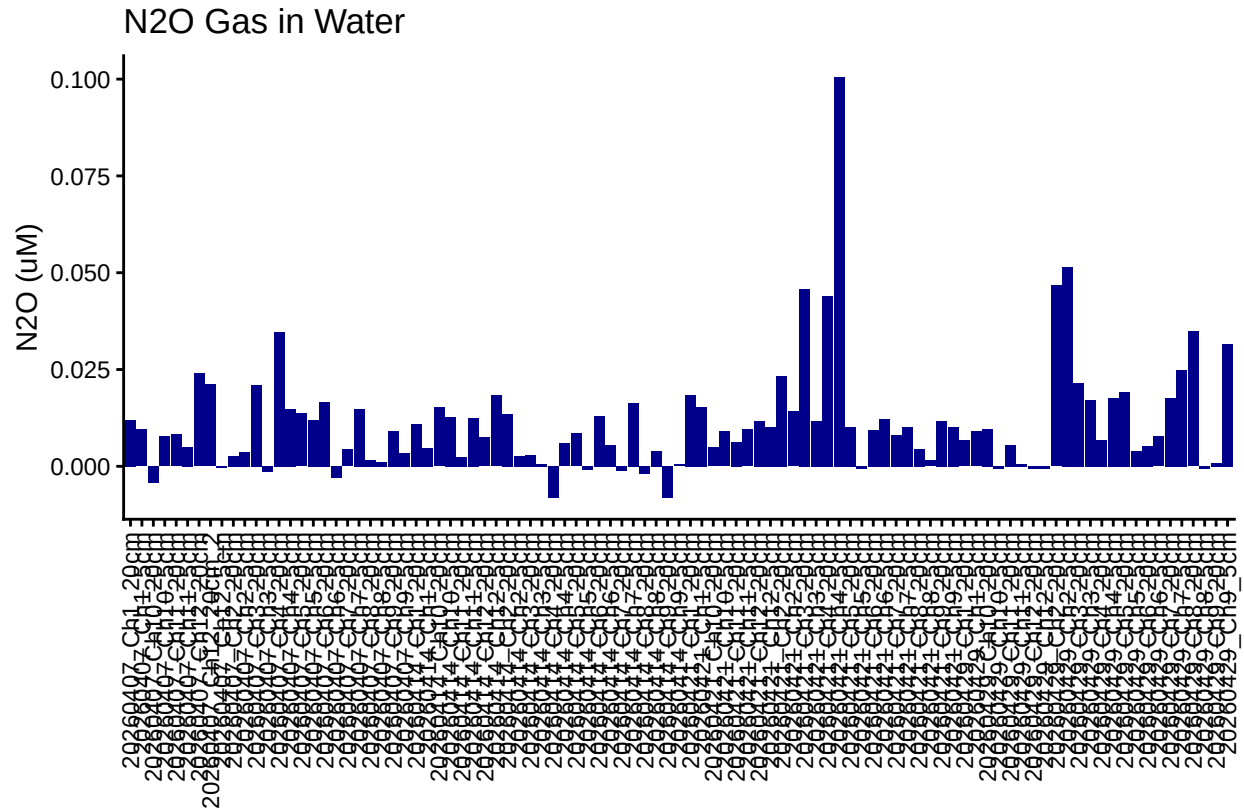


N2O: Green = Within Range (Not Dil Cor)



6 Calculate gas in water





7 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## Some sample IDs are missing from data.
```

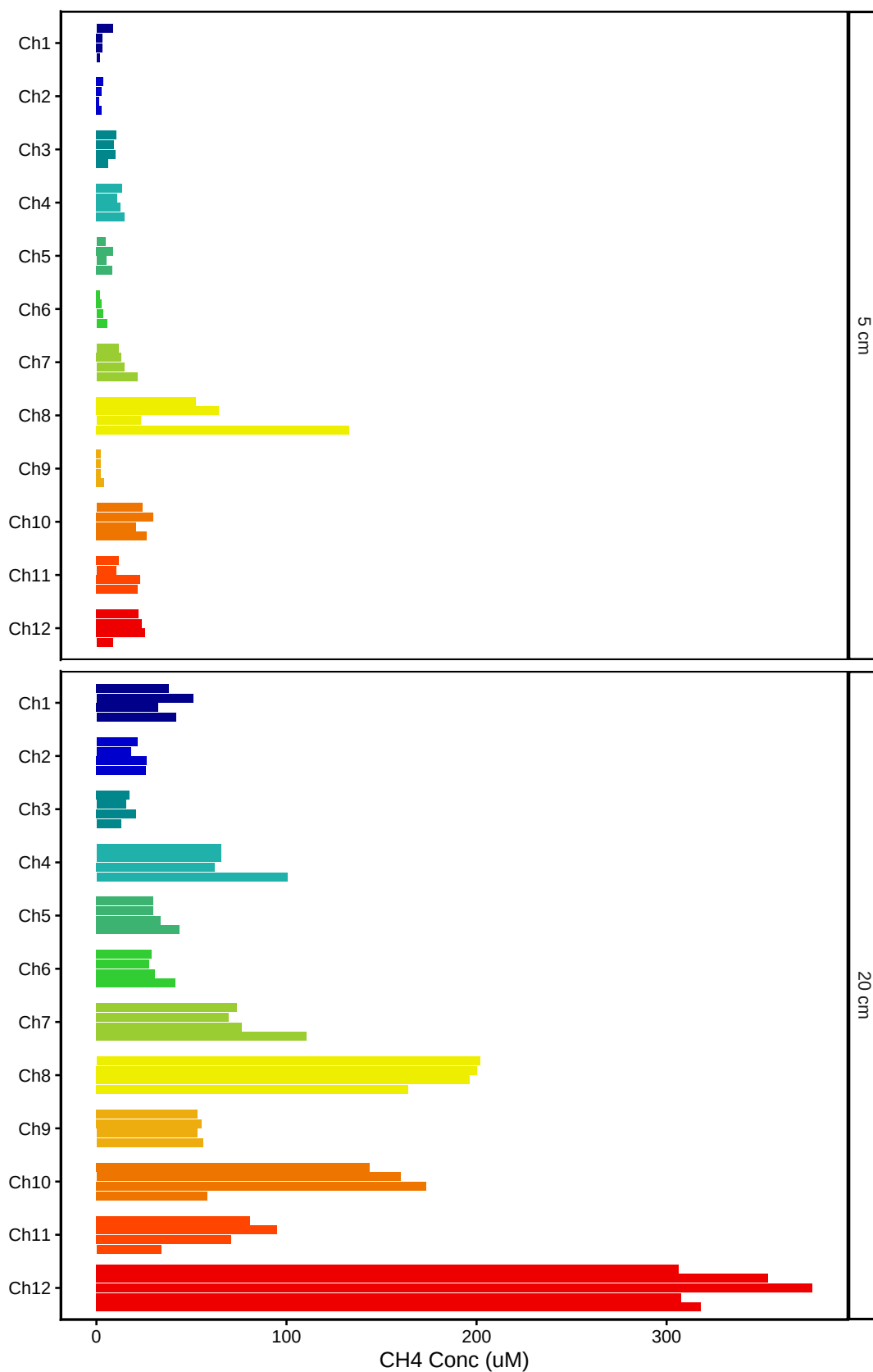
```
## [1] "Ch1-40" "Ch1-80" "Ch2-40" "Ch2-80" "Ch3-40" "Ch3-80" "Ch4-40"
## [8] "Ch4-80" "Ch5-40" "Ch5-80" "Ch6-40" "Ch6-80" "Ch7-40" "Ch7-80"
## [15] "Ch8-40" "Ch8-80" "Ch9-40" "Ch9-80" "Ch10-40" "Ch10-80" "Ch11-40"
## [22] "Ch11-80" "Ch12-40" "Ch12-80"
```

7.1 Organize Data

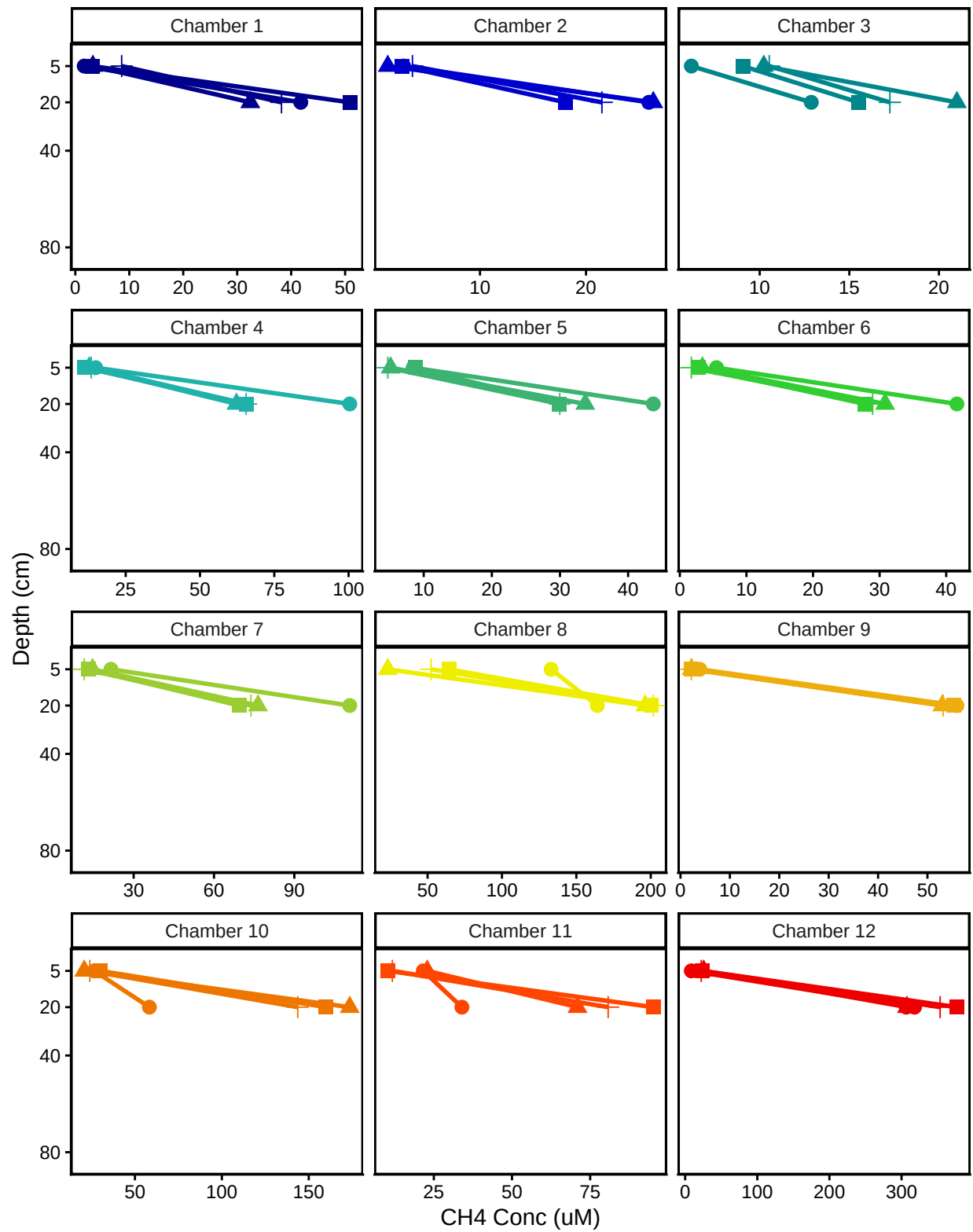
7.2 Visualize Data CH4

```
## Visualize Data
```

GENX: Porewater CH4



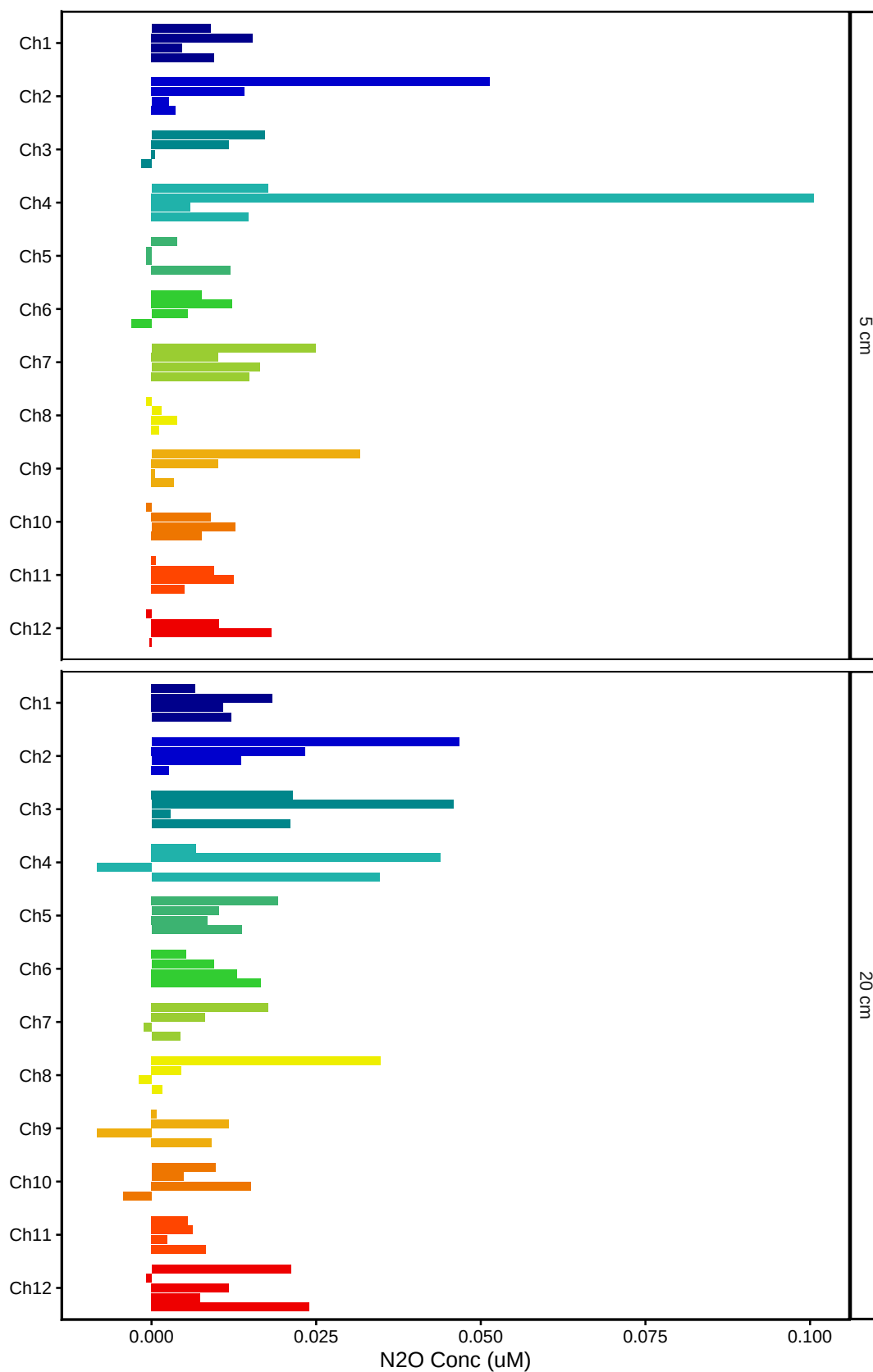
GENX: Porewater CH4 by Chamber



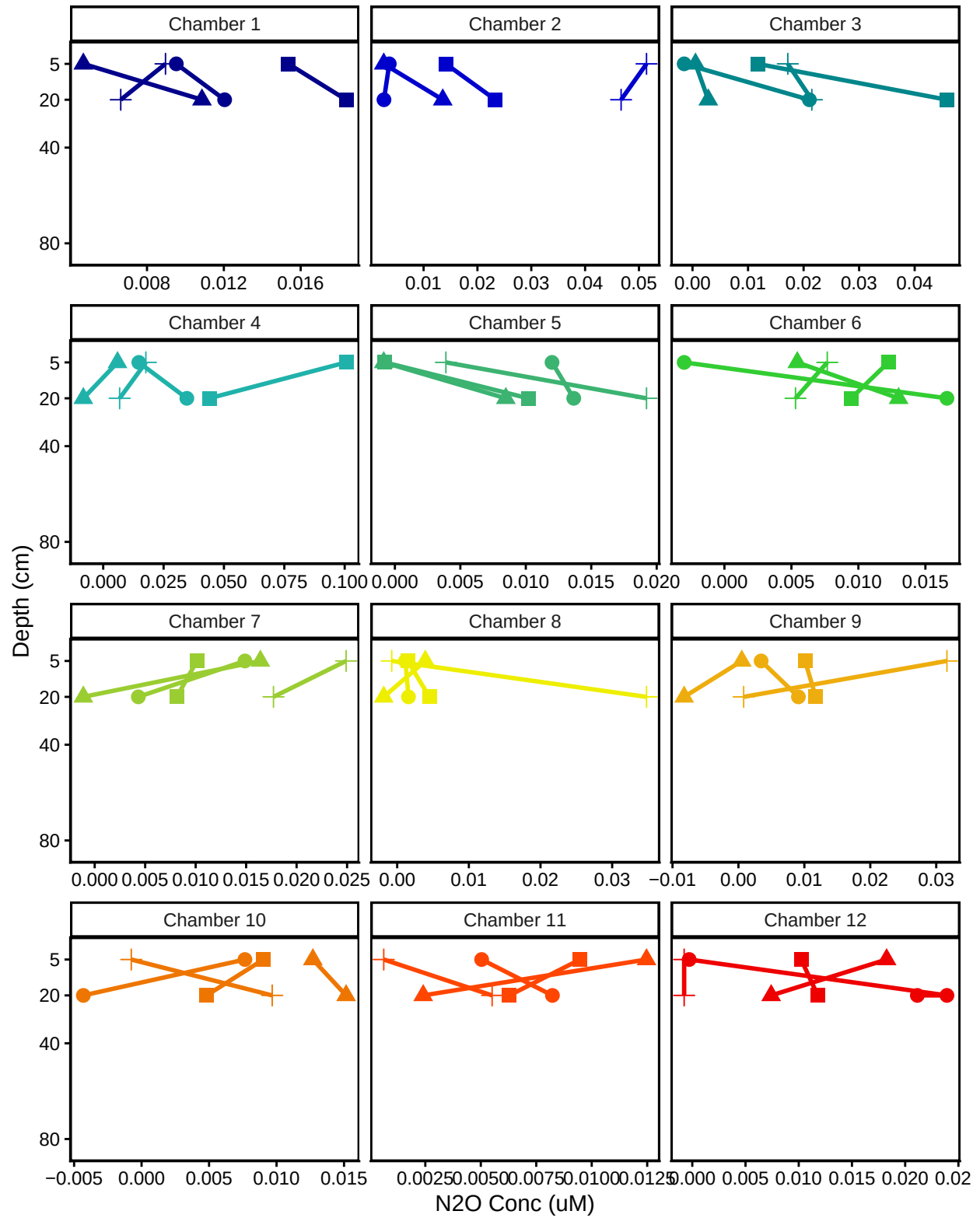
##Visualize Data N2O

```
## Visualize Data
```

GENX: Porewater N2O



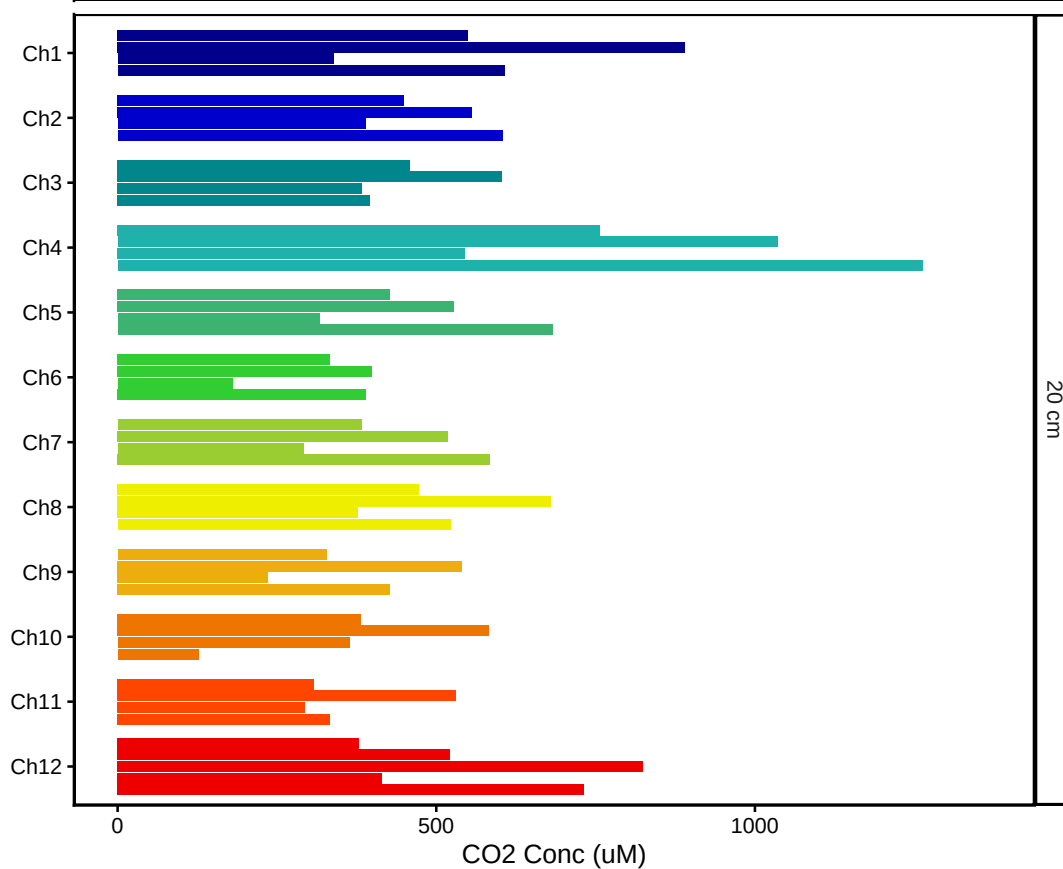
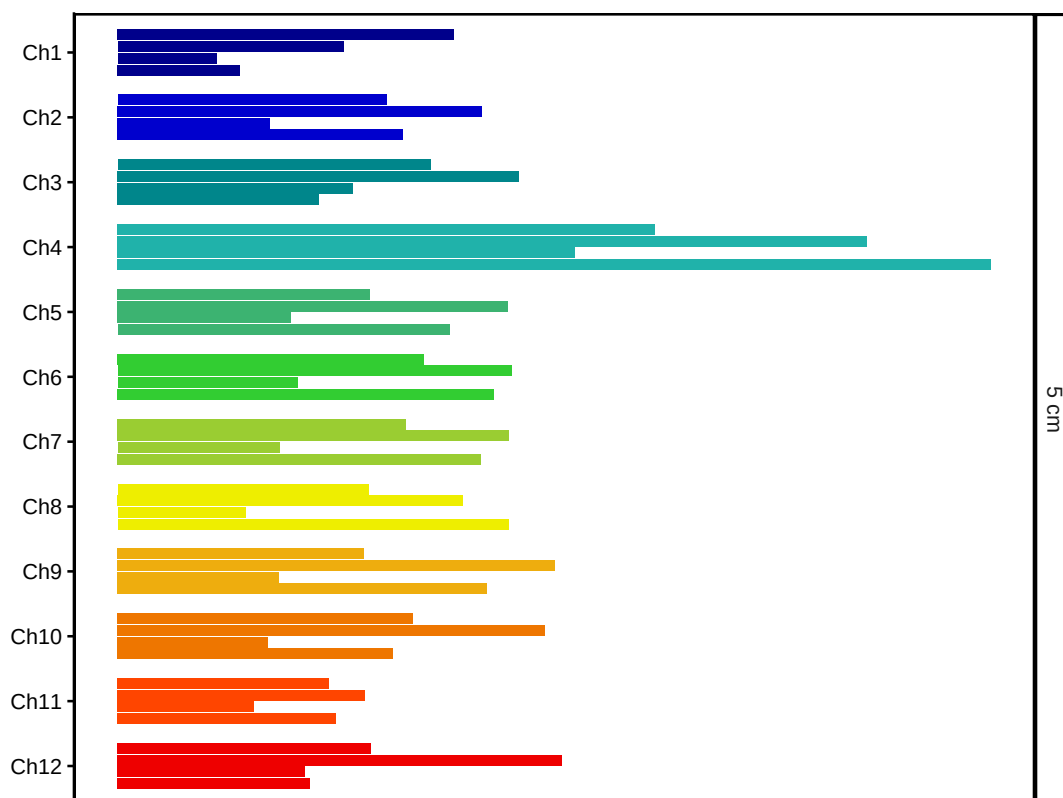
GENX: Porewater N2O by Chamber



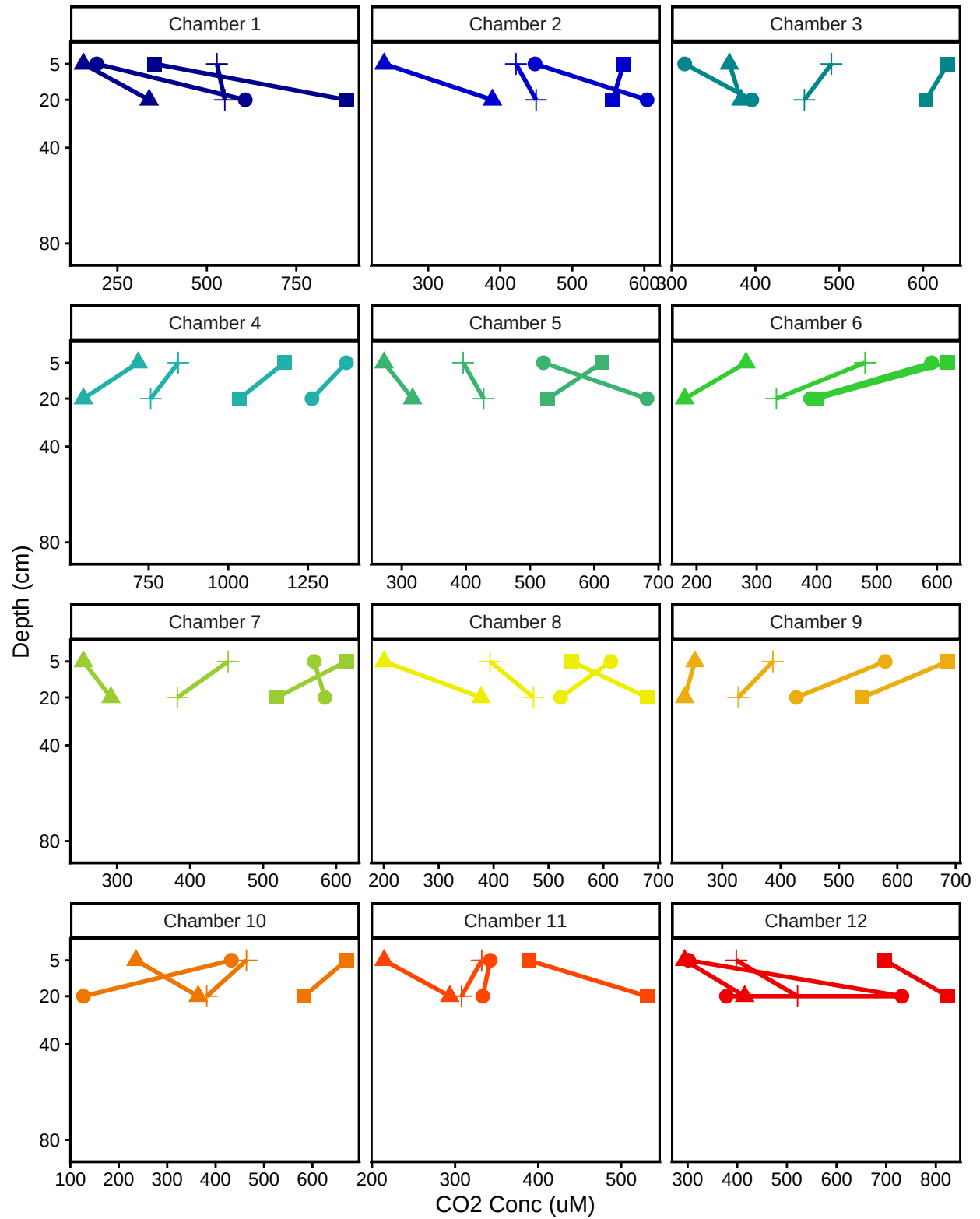
##Visualize Data CO2


```
## Visualize Data
```

GENX: Porewater CO2



GENX: Porewater CO2 by Chamber



##Export Data

#end